Jakub Kopal

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Computational neuroscientist translating machine learning research into clinical and applied practice.

Working experience

2024 – Present **University of Oslo, Oslo, Norway**, *MSCA Postdoctoral fellow*,
Supervisor: Ole A. Andreassen,
Marie Curie Postdoctoral fellowship
Coordinate AI research projects between University of Oslo and UC San Diego
Participate in Horizon Europe projects (environMENTAL, GenAI4EU, EU4Health - Brain AID)
Contribute to grant proposals for Horizon Europe, focusing on AI for healthcare
Lead cross-functional collaboration between computational and clinical consortium partners.
Deliver end-to-end analytical workflows for large-scale health datasets.

2021 – 2024 **McGill University & Mila - Quebec AI Institute, Montréal, Canada**, *Postdoctoral fellow*,
Supervisor: Danilo Bzdok, Co-supervisor: Sébastien Jacquemont,
Organized bi-weekly Journal Club. Mentored undergraduate and graduate students
Applied advanced machine learning methods to large-scale biomedical datasets
Led collaborations between Mila and CHU Sainte-Justine, translating clinical research questions into machine-learning pipelines and coordinating delivery across computational and medical teams.

2017 – 2021 **Czech Academy of Sciences, Prague, Czech Republic**, *Ph.D. student*,
Institute of Computer Science, Complex networks and brain dynamics group,
Developed computational models and statistical tools for analyzing brain connectivity
Applied advanced statistical modeling and network theory to neuroimaging data
Collaborated with clinics to translate research into interpretable medical findings.

Jun 2015 – Aug 2015 **University of Glasgow, Glasgow, UK**, *Internship*,
Department of Bioengineering,
Project: Development of a rapid point of care diagnostic device, utilizing surface acoustic waves to manipulate and then sense biological marker.

Education

2017 – 2021 **Brain and Cognition Research Center (CerCo), Toulouse, France**,
Doctoral studies in Neuroscience (cotutelle PhD with UCT Prague),
Supervisor: Emannuel J. Barbeau, Ph.D.

2016 – 2021 **University of Chemistry and Technology, Prague, Czech Republic**,
Doctoral studies in Technical cybernetics (cotutelle PhD with UPS Toulouse),
Research on computational modeling, signal processing, and machine learning in biomedicine.
Teaching: Applications of Computer Science (Czech and English language),
Supervisor: Oldrich Vysata M.D., Ph.D., Co-supervisor: Jaroslav Hlinka, Ph.D.

2013 – 2016 **University of Chemistry and Technology, Prague, Czech Republic**,
Master's degree,
Field of study: Applied engineering informatics,
Focus on optimization methods, process modeling, and applied machine learning.
Relevant coursework: Engineering Optimization, Numerical Methods, Mathematical Modeling of Processes, Neural Networks, Pattern Recognition, Discrete Event Simulation.

Sep 2014 – Dec 2014 **University of Oulu, Oulu, Finland**,
Erasmus exchange student,
Field of study: Bioengineering and computer science,
Project: Dynamic measurements of balance with footwear sensors for BoneWell Intelligence Ltd.

2010 – 2013 **University of Chemistry and Technology, Prague, Czech Republic**,
Bachelor's degree,
Field of study: Process engineering, informatics, management,
Strong foundation in chemical engineering and control systems.
Relevant coursework: Measuring and Control Engineering, Chemical Engineering, Signal Processing, Enterprise Process Management, Mathematics, Physics.
Bachelor thesis: Dynamic Parameters of EEG during Meditation.

Academic record

Google scholar **https://scholar.google.com/citations?user=GfLE4fgAAAAJ**.

Selected publications

The end game: Respecting major sources of population diversity,
Kopal, J., Uddin, L., Bzdok, D.,
Nature Methods.

A pattern-learning algorithm associates copy number variations with brain structure and behavioural variables in an adolescent population cohort,
Kopal, J., Jacquemont, S., Bzdok, D., et al.,
Nature Biomedical Engineering.

Rare CNVs and phenome-wide profiling highlight brain-structural divergence and phenotypical convergence,
Kopal, J., Jacquemont, S., Bzdok, D., et al.,
Nature Human Behavior.

Deep learning reveals that multidimensional social status drives population variation in 11,875 US participant cohort,
Marotta, J., Kopal, J., Bzdok, D, et al.,
PLOS One.

Leadership, Mentoring & Service

Grant contribution,
Horizon Europe grants: environMENTAL, Generative AI 4 EU
Marie Curie Postdoctoral Fellowship
Barrande Fellowship – PhD mobility grant between Czech Republic and France.

Community Building,
Organize department-wide workshops and social initiatives to foster cross-team collaboration.

Science communication,
Oral presentations at World Congress of Psychiatry, Nordic Conference on Future Health, World Congress of Psychiatric Genetics, and the Organization for Human Brain Mapping
Data visualization consultant, University of Oslo.

Supervision & mentoring,
Master and Bachelor students in computational neuroscience and machine learning.

Other activities,
Reviewer for NeuroImage, Scientific Reports, OHBM conference
Member of senior researcher hiring committees: University of Oslo, Mila - Quebec AI Institute.

Technical Skills

Development	Python (scikit-learn, PyTorch, pandas, numpy, JAX), R, MATLAB, bash | Git version control | Docker/Singularity containerization | HPC clusters (SLURM)
ML & Statistics	Supervised/unsupervised learning, Bayesian modeling (PyMC3), multivariate methods (PCA, CCA, PLS), time-frequency analysis, tensor decomposition (PARAFAC2)

Applied AI

Translating domain problems into machine learning formulations

Coordinating international research projects across academic, clinical, and industry partners

Developing reproducible analytics pipelines for large-scale biomedical data

Communicating findings and trade-offs to non-technical stakeholders

International training courses

Nov 2015 **Universidad Politecnica de Madrid, Madrid, Spain**,
Athens exchange program: Biosignal processing.

Apr 2014 **Technische Universitat Munich, Munich, Germany**,
Athens exchange program: Manipulation of time series in time and frequency spectrum.

Nov 2013 **Ecole Nationale Superieure de Techniques Avancees, Paris, France**,
Athens exchange program: Medical Imaging.

Languages

Czech	Native Speaker	
English	Fluent (C1+)	*+ 3 semesters of Academic writing course*
French	Fluent (C1)	
Norwegian	B1, actively improving	

References available upon request.